Cytoplasm is thick, aqueous, semi-transparent and elastic. The major structures in the cytoplasm of prokaryotes are nucleoid (containing DNA), ribosomes and reserve deposits called inclusions. Prokaryotic cytoplasm lacks certain features of eukaryotic cytoplasm such as a cytoskeleton and cytoplasmic streaming.

Ribosomes

All living cells contain ribosomes. They are the sites of protein synthesis. High number of ribosomes represents the high rate of protein synthesis. Prokaryotic ribosomes are freely found in the cytoplasm, whereas eukaryotic ribosomes are attached to the cell membrane. Prokaryotic ribosomes consists of protein and a type of RNA called ribosomal RNA. They are smaller and less dense than the eukaryotic ribosomes. The ribosomes of prokaryotes are 70S where as that of eukaryote are 80S (Figure 7.12).

Nucleus

The nuclear area has the hereditary material of most bacteria. It contains a single, circular, long, continuous, thread like double stranded DNA called the bacterial chromosome. Some bacteria with linear chromosome also exist. It carries the information required for the cells structure and function. They are not surrounded by a nuclear envelope and are devoid of highly conserved histone proteins. The nuclear area can be spherical, elongated or dumbbell shaped. In actively growing bacteria, as much as 20% of the cell volume is occupied by DNA, because such cells presynthesize nuclear material for future cells. The chromosome is attached to the cell membrane. Proteins in the plasma membrane are believed to be responsible for the replication of the DNA and segregation of the new chromosomes to daughter cells in cell division.

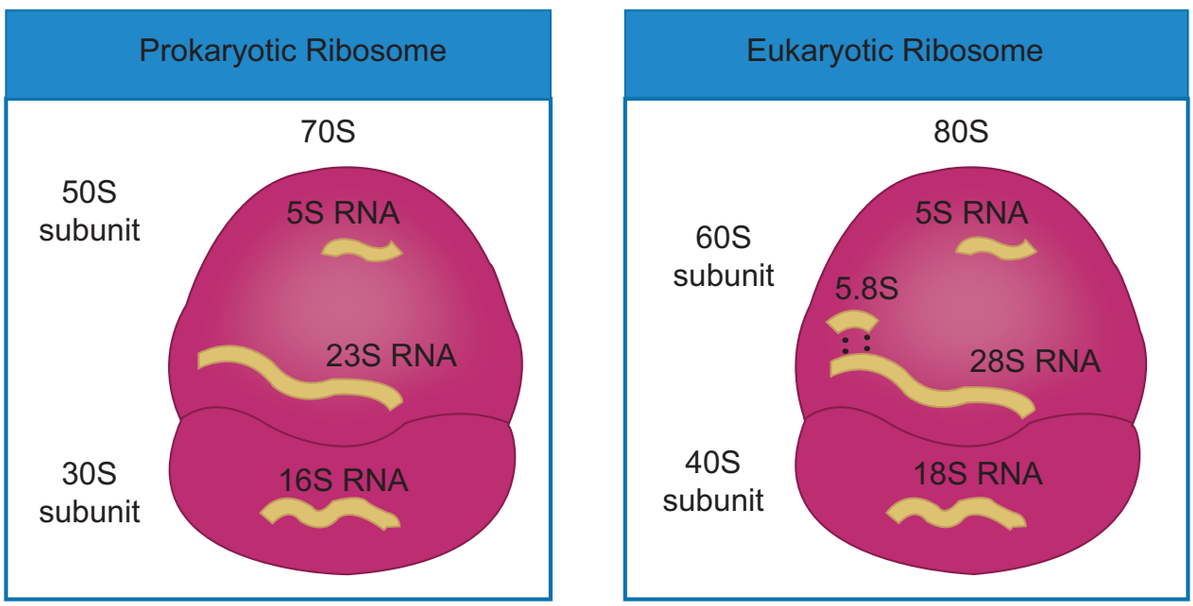


Figure 7.12: Prokaryotic and Eukaryotic Ribosomes

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